# Voltiva 1.0

Creative Design Project

BECS 22811


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## Project Objectives:

**Monitor Electrical Parameters:** Develop a system to continuously monitor electrical parameters (current and voltage) using IoT to improve real-time oversight and control.

**Remote Access and Control:** Provide users with remote access via a and mobile application to view electrical data and control connected devices.

**Fault Detection and Alerts:** Enable real-time detection of electrical faults (like overcurrent or voltage fluctuations) and send alerts through the application to prevent equipment damage.

**Energy Efficiency Insight:** Offer insights into energy consumption patterns, helping users make informed decisions to enhance energy efficiency.

## Problem Statement:

Many users face challenges in monitoring their electrical systems in real-time, particularly in detecting faults early or identifying inefficient energy consumption. Existing solutions are often too expensive or complex for personal or small-scale use. There is a need for an affordable, user-friendly IoT-based system to monitor electrical parameters in real-time, offering remote control and fault alerts. This project aims to bridge the gap by creating a scalable, accessible system that provides critical electrical data to users, helping improve safety and energy management.

## Literature Review:

**IoT in Electrical Monitoring:** Research on existing IoT applications in monitoring electrical parameters shows that integrating sensors with microcontrollers (like ESP32) allows real-time data collection and transmission, providing users with immediate feedback on power usage and potential risks.

**ESP32 for IoT Applications:** The ESP32 microcontroller, with built-in Wi-Fi capabilities, is widely used in IoT applications due to its affordability, efficiency, and ability to process and transmit data over Wi-Fi networks. Studies demonstrate its reliability in managing IoT data communication.

**Voltage and Current Sensors:** The ACS712 current sensor and ZMPT101B voltage sensor are commonly used in similar projects due to their accuracy and compatibility with low-power devices. These sensors enable precise measurement of electrical parameters.

**Cloud and Real-time Data Access:** Firebase provides a simple and reliable way to store real-time data and is well-integrated with web development tools like Node.js and React. This setup allows seamless real-time access to data, enhancing the usability of IoT monitoring systems.

**User Application for Control:** Studies highlight that React and Node.js are effective for building cross-platform, interactive applications, offering users a way to access and control systems remotely.

## Scope:

**Domain:** The project falls under the domain of IoT-based electrical monitoring and control, focusing on home or small-scale applications.

**Features:**

**Real-time Data Monitoring:** Display live electrical parameters, such as current and voltage, accessible via a mobile application.

**Remote Control of Connected Devices:** Control devices through the solid-state relay, enabling switching on/off remotely.

**Data Logging and Visualization:** Record and visualize historical data on Firebase for easy analysis of energy consumption patterns.

**Fault Alerts:** Provide immediate notifications through the app when electrical faults (like overcurrent) are detected.

**User-friendly Interface:** Develop an intuitive mobile application using React and Node.js for easy interaction.

## Methodology:

**Phase 1: Planning and Research**

**Problem Identification:** Assess the specific need for real-time monitoring and potential fault detection in electrical systems.

**Literature Review:** Study existing research and similar IoT projects to understand effective designs, methods, and tools.

**Solution Exploration:** Identify solutions for remote monitoring, control, and data processing using ESP32 and relevant sensors.

**Budget Estimation:** Determine the cost of components based on finalized specifications.

## Phase 2: Design and Simulation

**Circuit Blueprint Design:** Design the electrical circuit schematic, including the ACS712 current sensor, ZMPT101B voltage sensor, and solid-state relay.

**Simulation:** Use simulation tools to test the circuit design and troubleshoot preliminary issues before physical assembly.

**Component Finalization and Purchase:** Finalize the list of apparatus based on simulation outcomes, and purchase required components.

## Phase 3: Development

**Prototype Assembly:** Assemble the basic prototype and troubleshoot initial hardware issues.

**Microcontroller Programmin**g: Write and upload code to ESP32 for sensor data collection, processing, and relay control.

**Mobile Application Development:** Develop the frontend (React) and backend (Node.js with Firebase integration) of the application, allowing users to view and control the system remotely.

**Data Processing and Storage:** Use Firebase to store real-time and historical sensor data for easy access and retrieval in the mobile application.

### Phase 4: Testing and Optimization

**Hardware Testing and Optimization**: Conduct tests on sensor accuracy and relay response to ensure reliability under various conditions.

**Software Testing:** Perform rigorous testing on the application for smooth interaction, data display, and control features.

**Fault Handling Mechanism:** Program fault alerts and test the system's response to electrical abnormalities.

**Phase 5: Finalization and Implementation**

**Hardware Finalization:** Secure all components and ensure durability for continuous use.

**Software Finalization**: Finalize application features, focusing on UI/UX for ease of use.

**Final Testing:** Perform full-system testing to ensure smooth integration and functionality.

**Project Report:** Document each phase, challenges, and outcomes to prepare the final project report.

# Estimated Budget of the Project

| Component Name | Quantity | Price (Rs.) | Total Price (Rs.) |
|---|---|---|---|
| ESP32 Microcontroller | 1 | 1600 | 1600 |
| ACS712 current sensor | 1 | 540 | 540 |
| ZMPT101B voltage sensor | 1 | 550 | 550 |
| Relay Module (Solid-State Relay) | 1 | 150 | 150 |
| HLK-PM03 Power Supply Module | 1 | 860 | 860 |
| Extra | | 3000 | 3000 |
| **Total Cost** | | | **6700** |

# Complete Gantt Chart of the Project

| Task | Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6 | Week 7 | Week 8 | Week 9 | Week 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Problem Identification | | | | | | | | | | |
| Literature Review | | | | | | | | | | |
| Finding solutions for the problem | | | | | | | | | | |
| Budget Estimating | | | | | | | | | | |
| Project Proposal preparation | | | | | | | | | | |
| Circuit Blue Print Designing | | | | | | | | | | |
| Circuit Simulation | | | | | | | | | | |
| Finalize the apparatus list according to final design | | | | | | | | | | |
| Purchasing Apparatus | | | | | | | | | | |
| Studying Programming | | | | | | | | | | |
| Basic prototype Building | | | | | | | | | | |
| Basic prototype Troubleshooting and Optimization | | | | | | | | | | |
| Microcontroller Programming and Sensor Data Processing | | | | | | | | | | |
| Mobile application Development (Both Frontend and Backend) | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|
| Hardware Troubleshooting and Optimization | | | | | | | | | | |
| Software Testing | | | | | | | | | | |
| Finalizing Hardware Construction | | | | | | | | | | |
| Finalizing Software Construction | | | | | | | | | | |
| Final Testing of the Device | | | | | | | | | | |
| Final Implementation | | | | | | | | | | |
| Preparation of the Final Project Report | | | | | | | | | | |

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